Current, LEDs and resistors

IoT · Level 2 · Lesson 1



Today we will

- Explain that current flows only round a closed loop, pushed by the battery's voltage.
- Light an LED the right way round, with a resistor to limit the current.
- Use Ohm's law to choose a resistor, and spot a short circuit.

Loop, LED, resistor

- Current needs a closed loop
- The LED's long leg goes to +
- A resistor limits the current
- $V = I \times R$

Conductor hunt

1. Warm-up (5 min): Show a torch that won't light. Ask why.
2. Teach (8 min): The loop, conductors and insulators, the LED's resistor.
3. Build (7 min): Groups make a tester from batteries, a bulb and wires.
4. Test (15 min): Predict, then test ten objects; sort them on the board.
5. Wrap-up (5 min): Why is a wire copper inside plastic?

Before class, build and test one tester yourself. Use batteries only. Keep button batteries away from young children, and don't let pupils hold a bare wire across a battery: it gets hot.

Let’s discuss

1. Which result surprised you, and why?
2. Why does an LED need a resistor when a torch bulb doesn't?
3. Where could a short circuit happen by accident at school or at home?

Exit ticket

- Name one conductor, one insulator, and one safety rule for circuits.